

Elliott Schmechel

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SUMMARY

Computer Science student focused on backend, systems, and developer tooling. Recent projects include a 4-replica Kubernetes GPU training platform on Lambda Labs H100s, a serverless tutoring backend on Cloudflare Workers with a 21-tool MCP server, and a Proxmox-based homelab for infrastructure experimentation. Comfortable across Go, C++, Python, and TypeScript.

TECHNICAL SKILLS

Languages: Go, C++, Python, TypeScript, SQL, Bash, HTML/CSS
Backend / Data: REST APIs, Hono, FastAPI, Next.js, Drizzle ORM, PostgreSQL
Infrastructure: Docker, Kubernetes, Proxmox, Cloudflare Workers/Pages/D1, GitHub Actions, Lambda Labs
Linux / Systems: Arch Linux (daily driver), Ubuntu, Debian, self-hosted services, MCP servers, Wrangler
Libraries / Tools: React, Leaflet, Deepgram, paramiko, httpx, Cheerio, Express

PROJECTS

- DataForAll** (HTC 2026 Infrastructure Lead) | *Python, FastAPI, Lambda Labs GPU, Kubernetes, Docker* Feb 2026
- Led infrastructure for a distributed GPU training platform, provisioning H100 instances via Lambda Labs API with SSH-based container deployment
 - Built Kubernetes manifests for a 4-replica backend with secrets management, container registry, and multi-region load balancing
 - Implemented GPU worker lifecycle management with heartbeat monitoring, stale-instance detection, auto-cleanup, and cost estimation (\$3.29/hr per H100)
 - Integrated S3 object storage, PostgreSQL with Alembic migrations, and FastAPI services with training pipelines and LoRA fine-tuning
- LearnLM** (XHacks 2026 – Best Use of SFUCoursesAPI) | *TypeScript, Hono, Cloudflare Workers, React* Jan 2026
- Built the serverless backend on Cloudflare Workers with Hono, authoring 66% of commits across 6 D1 tables, 3-tier rate limiting, and a Vectorize RAG pipeline indexing 998 SFU courses
 - Built a Model Context Protocol (MCP) server exposing 21 tools for document critique, course search, and tutoring, integrated with VS Code, Cursor, and Claude Desktop
 - Implemented real-time voice tutoring via WebSocket Durable Objects with Deepgram STT/TTS, reducing response latency by switching to Llama 3.1 8B at the edge
 - Configured GitHub Actions CI/CD for zero-downtime deploys to Cloudflare Workers and Pages; continuing development as TutorNexus
- Beeptd** (JourneyHacks 2026 – Lone Wanderer Award) | *TypeScript, Hono, Drizzle ORM, D1, Leaflet* Jan 2026
- Solo-built a real-time proximity radar app end-to-end in 12 hours, designing 6 Drizzle/D1 tables with indexed geospatial queries using the Haversine formula
 - Implemented device-secret authentication, friend request flows (send/accept/reject/block), and location updates with automatic 24-hour TTL expiry
 - Built an interactive Leaflet map with marker clustering and a radar view, supporting friends-only and public proximity scopes within configurable radii
- Home Lab Infrastructure** | *Proxmox, Arch Linux, Ubuntu, Debian, self-hosted* Ongoing
- Run a multi-VM Proxmox homelab with VLAN isolation and Kubernetes experimentation across nodes, deploying self-hosted services including MCP servers and Obsidian sync
 - Daily-drive Arch Linux; build custom CLI tools and scripts to fill gaps in available software; configured local STT via whisper (gpt-oss) for offline transcription

EXPERIENCE

- Volunteer Fullstack Developer** Oct. 2025 – Present
United Nations Association - Vancouver Hybrid
- Collaborated with design and product teams to architect features and define technical scope for the website rewrite
 - Developed and integrated 15+ responsive React components from Figma specifications, improving page load speed by 25%
 - Deployed application on Cloudflare Workers/Pages with 99.9% uptime, reducing hosting costs by \$12/month vs prior hosting

EDUCATION

Langara College Vancouver, BC
Associate of Science in Computer Science Sep. 2024 – Dec 2026